CARESETU AI

Autonomous Inpatient Discharge Orchestration on ServiceNow

Connecting fragmented department silos to eliminate discharge delays with grounded AI, deterministic urgency, and human-approved action.

Zero-Mutation AI Boundary

- AI never mutates databases autonomously
- Pre-approved proposals require authorized staff sign-off
- Server-side state revalidation & idempotency guards
- Strict non-clinical operational boundary

Deterministic Urgency

- Transparent 0–100 integer operational risk score
- Linear additive delay clock with category deduplication
- Separates urgency priority from timeline projection
- 100% explainable & audit-ready logic

Native ServiceNow Architecture

- Scoped application: x_caresetu_ai
- React 18 SPA embedded in UI Page
- Scoped REST APIs, ACLs, and RBAC security
- Scheduled chasers for TPA & shift handoff SLAs

PROJECT METADATA: ServiceNow Hackathon 2026 | Application Scope: x_caresetu_ai | Instance: act-hck-4161-0029.lab.service-now.com

THE HUMAN REALITY OF DISCHARGE DELAYS

Eight Hours in a Post-Operative Hallway

The problem in hospitals today isn't clinical care. It is broken operational coordination.

The Personal Genesis

- My grandfather (Nanaji) recently underwent open-heart surgery.
- At 10:00 AM, his surgeon completed rounds and declared him medically ready for discharge.
- His vitals were stable, his surgical wound was healing, and he was eager to return home to his family.
- Yet, he sat in a recovery chair in a drafty hallway until 6:00 PM — eight hours after clinical clearance.
- He wasn't waiting on medical care. He was waiting on paperwork, transport, and insurance phone calls.
- Every nurse was exhausted, every family member was anxious, and nobody had end-to-end visibility.

The Silent Hospital Cascades

- TPA Insurance clearance sat unworked in a shared email inbox.
- Cardiology physical discharge summary was waiting on a shift change.
- Post-acute durable medical equipment delivery was never dispatched.
- Downstairs in the Emergency Department, critical cardiac patients were held on gurneys because Nanaji's bed was registered as 'occupied'.
- This delay did not stem from a lack of dedication by medical staff.
- It stemmed from disconnected systems, unowned operational blockers, and absence of proactive follow-up.

THE CORE LESSON: Hospitals don't lack heroic clinicians. They lack closed-loop operational coordination across department silos.

THE SYSTEMIC PROBLEM

Why Discharge Delays Paralyze Modern Hospitals

Department silos, invisible bottlenecks, and absence of accountability create hospital gridlock.

1. Fragmented Silos

- Nursing, Pharmacy, Transport, Billing, and Insurers operate in disconnected tools.
- No single person or dashboard sees the cross-department dependency chain.
- Communication relies on ad-hoc phone calls and hallway sticky notes.
- Status updates occur only when an angry family member inquires.

2. Invisible Blockers

- Operational deadlines pass silently without automated alerts.
- TPA/Insurance queries sit unanswered for hours without follow-up timers.
- Shift handoffs go unacknowledged across department transitions.
- Delay only surfaces when the patient is already past promised departure.

3. Cascading Gridlock

- Discharge delays trigger Emergency Department boarding and hallway admissions.
- Surgeons cancel scheduled procedures due to lack of inpatient bed availability.
- Severe staff fatigue and nurse burnout from manual chasing.
- In a 400-bed hospital, a 3-hour delay wastes ~1,200 bed-hours every single day.

KEY INSIGHT: Discharge coordination is not a clinical diagnosis problem — it is an enterprise workflow and service orchestration problem.

THE CARESETU PARADIGM

Closed-Loop Operational Orchestration on ServiceNow

Four grounded architectural principles transforming hospital workflow.

1. Explainable Deterministic Urgency

- Transparent 0–100 integer operational risk rules.
- Linear additive timeline clock with category deduplication.
- Zero black-box guessing: every point maps to a code-backed blocker.
- Separates priority urgency from projected delay minutes.

2. Grounded AI Briefings

- AI Care Coordinator aggregates live ServiceNow table records.
- Answers natural language queries strictly using in-scope table state.
- Never hallucinates clinical diagnoses or unsupported operational claims.
- Fluent Now Assist Skill ready; deterministic fallback active.

3. Fix It For Me (Human-Approved)

- AI formulates read-only proposals for allowlisted operational fixes.
- Zero autonomous database mutation: targets remain 100% untouched.
- Authorized staff reviews rationale and clicks 'Approve & Execute'.
- Server authorization check, target revalidation & idempotency lock.

4. Autonomous Background Schedulers

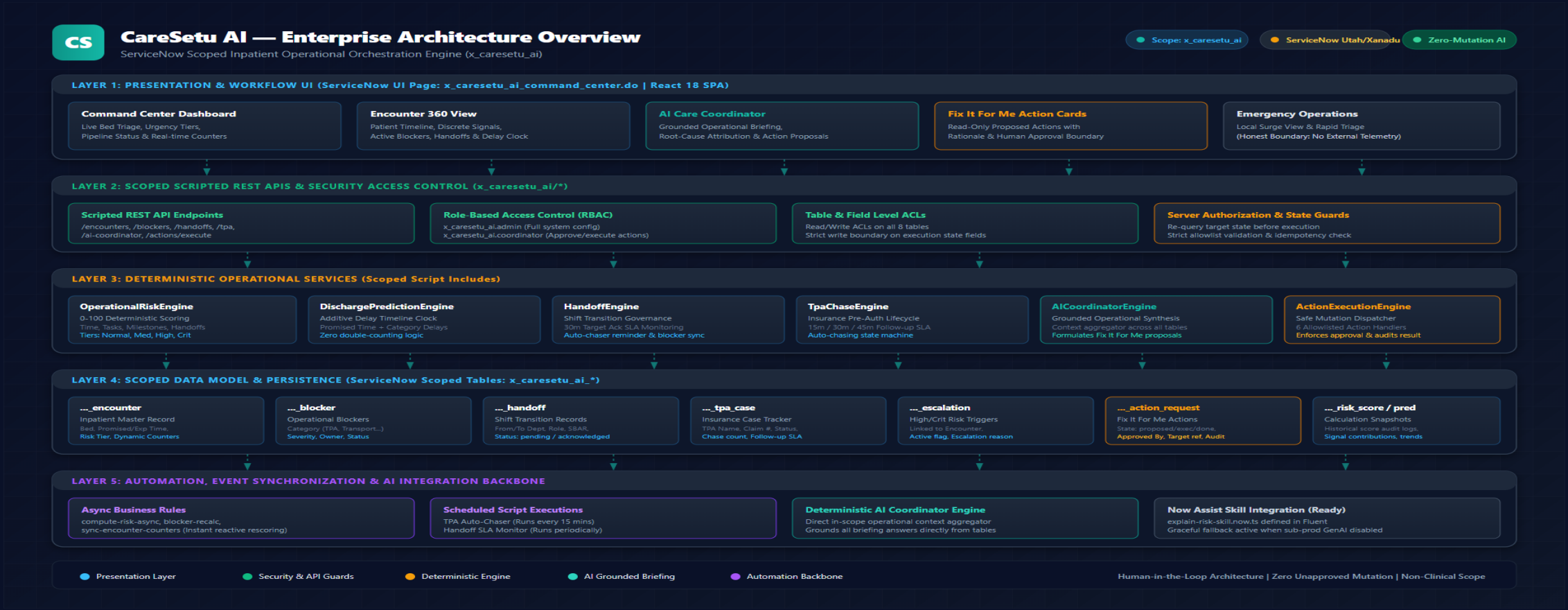
- TPA Auto-Chaser executes every 15m to monitor insurance pre-auth SLAs.
- HandoffEngine tracks 30-minute shift acknowledgment deadlines.
- Async business rules recalculate risk & prediction on any data change.
- Automatic escalation record creation when risk crosses High tier (>= 60).

PLATFORM FIT: ServiceNow is the ideal System of Action: providing auditable records, enterprise RBAC, and deterministic event engines.

SYSTEM ARCHITECTURE

End-to-End Enterprise Architecture Overview

5-layer scoped ServiceNow stack: Presentation, Security APIs, Operational Engines, Data Model, Automation.

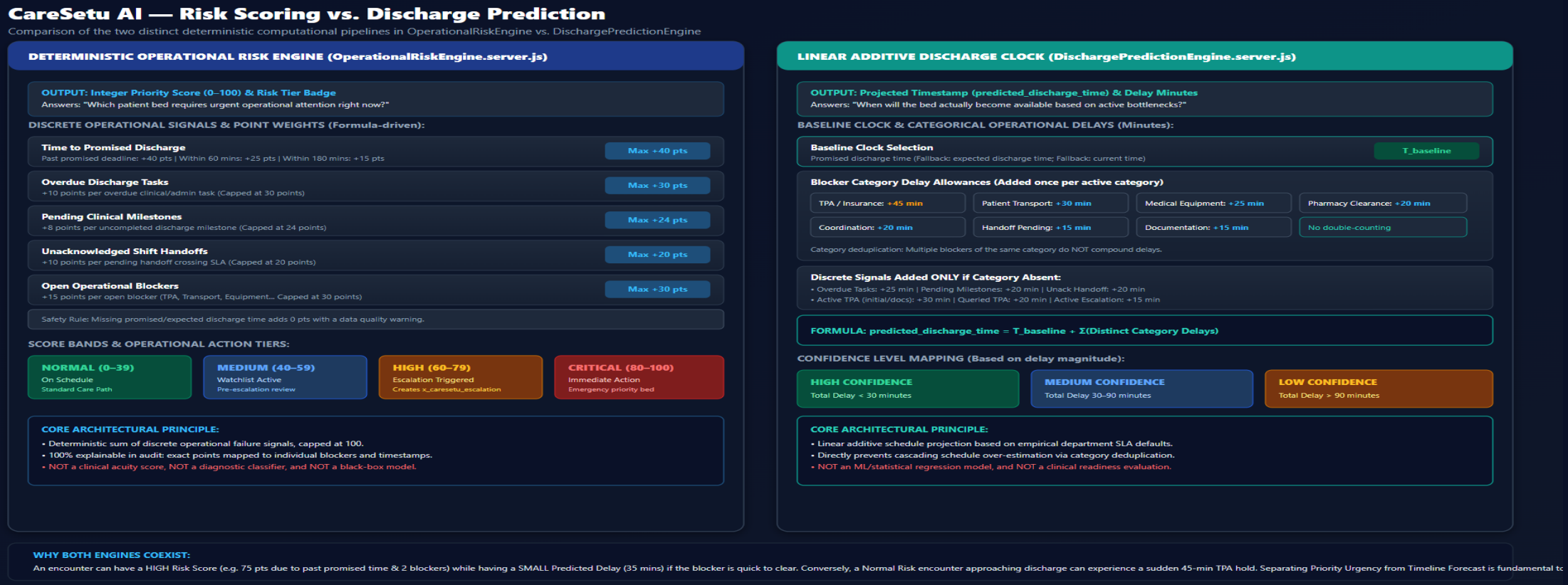


LAYER SUMMARY: Presentation: React 18 SPA in UI Page | Security: Scoped REST APIs & ACLs | Engines: 6 Script Includes | Data: 8 Scoped Tables

COMPUTATIONAL LOGIC

Deterministic Risk Scoring vs. Discharge Prediction

Separating Operational Priority Urgency (0–100) from Delay Timeline Clock (+Minutes).

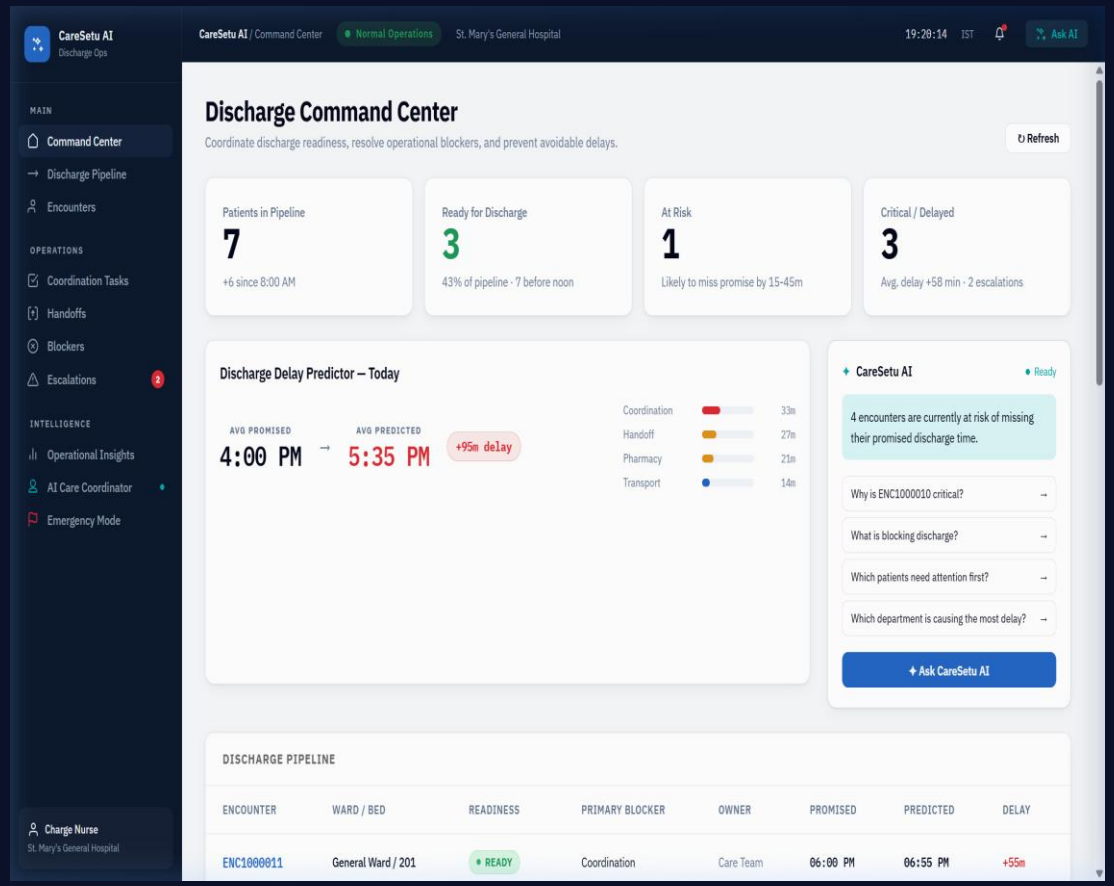


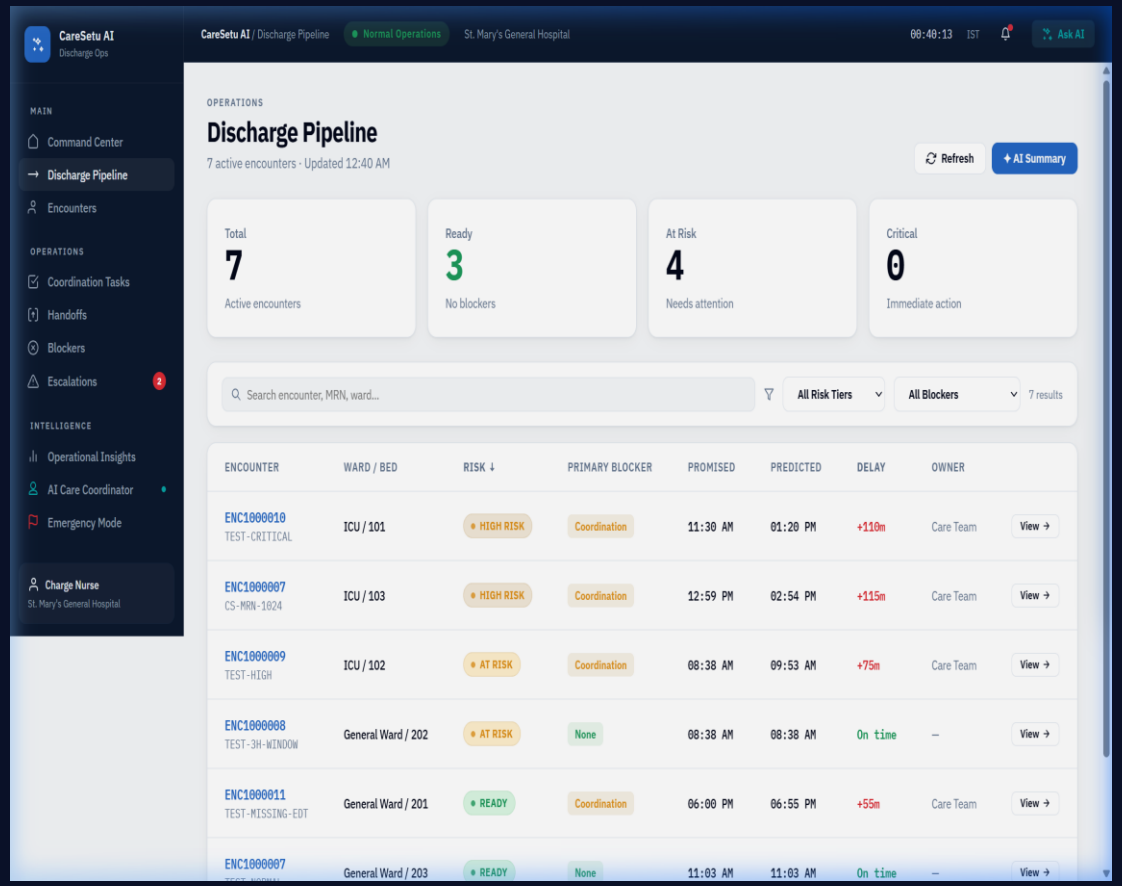
CODE-BACKED LOGIC: Deterministic integer score (0–100) drives triage urgency; Linear additive clock (+Minutes) drives schedule transparency. Neither is an ML model.

LIVE PLATFORM DEMONSTRATION

Command Center & Discharge Pipeline

Real-time bed triage, urgency badges, and bottleneck visibility across all active inpatient beds.



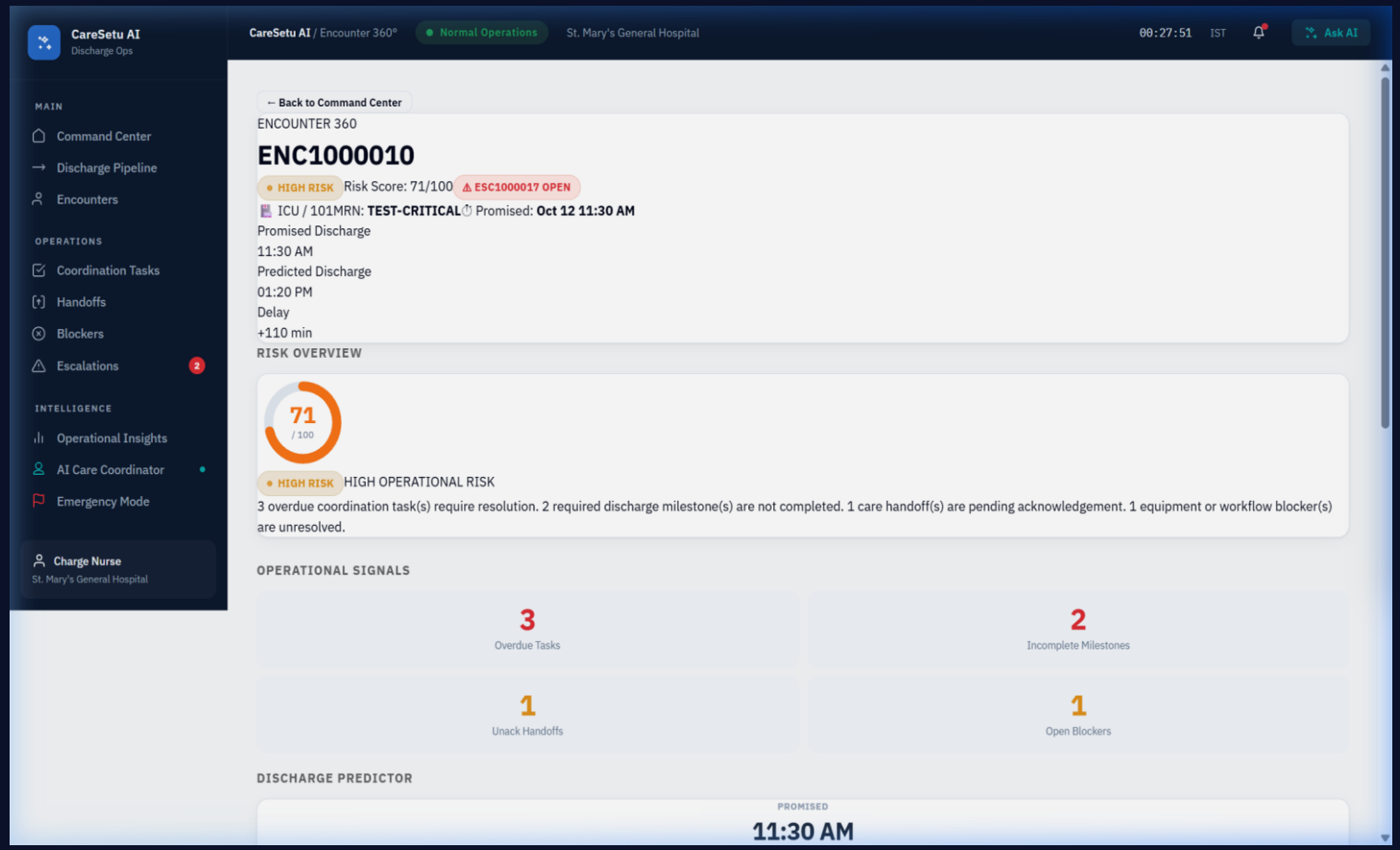


LIVE SYSTEM: Real live application captures: Left = Command Center Dashboard with Triage Badges | Right = Pipeline View by Stage

PATIENT DEEP-DIVE

Encounter 360: Full Operational Telemetry

Comprehensive 360-degree visibility into blockers, clinical milestones, shift handoffs, and delay clock.



Encounter 360 Signals

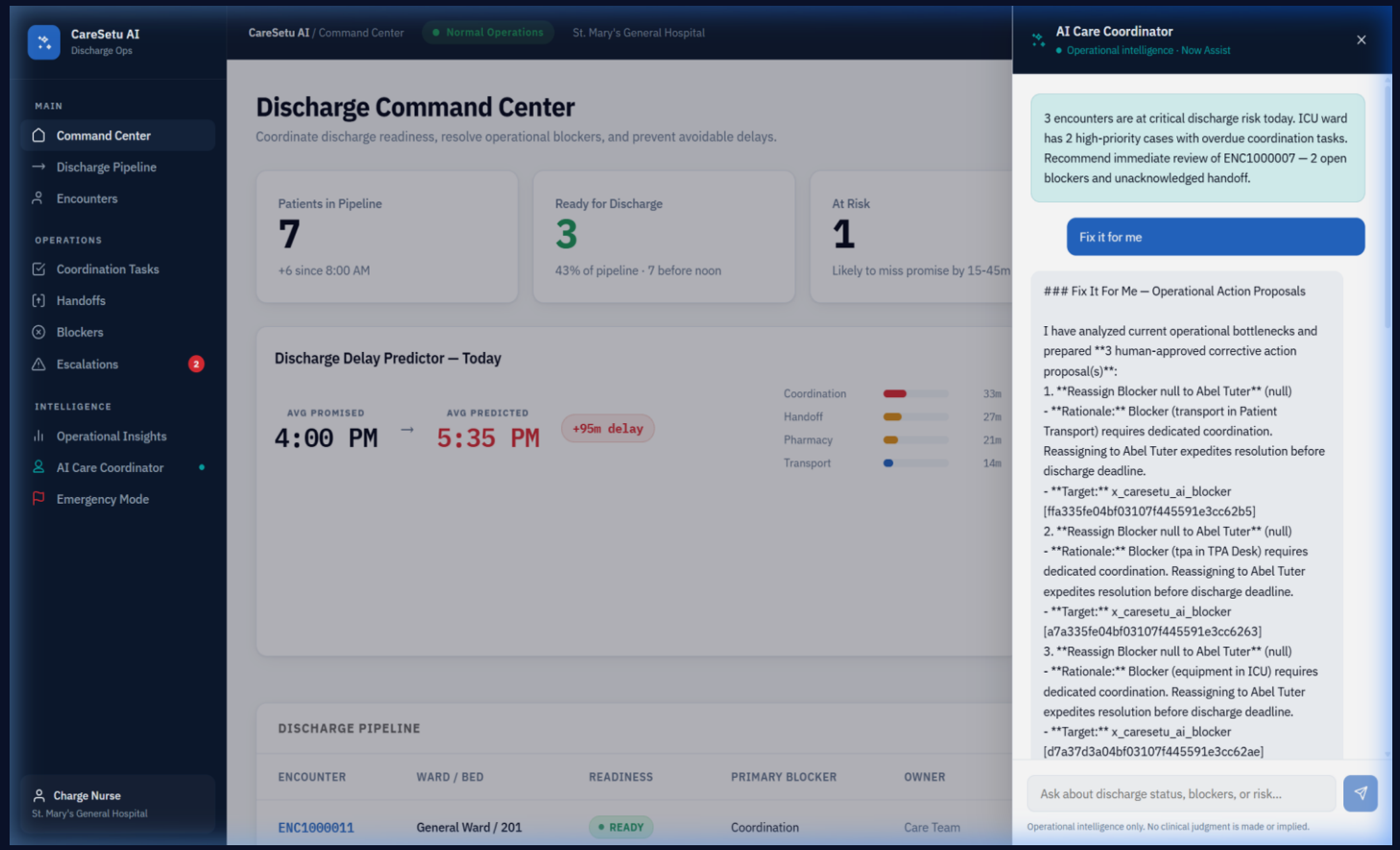
- Patient Context: Synthetic test record ENC1000007 (Bed 304)
- Promised Discharge Deadline vs. Projected Delay Clock (+45m)
- Discrete Signal Breakdown: Exact points contributing to 75 Risk Score
- Active Blockers Tab: TPA Pre-auth delay categorized with owner & SLA
- Milestones Tab: Cardiology sign-off vs. post-acute equipment
- Shift Handoffs Tab: Sender, receiver, SBAR summary, and ACK countdown timer
- Zero PHI: Strictly synthetic data for compliant testing

TELEMETRY: Encounter 360 consolidates fragmented department telemetry into a single, real-time operational pane.

CONVERSATIONAL INTELLIGENCE

AI Care Coordinator: Grounded Operational Briefings

Instant, plain-language operational summaries grounded directly in ServiceNow tables without hallucinations.



Grounding & Safety

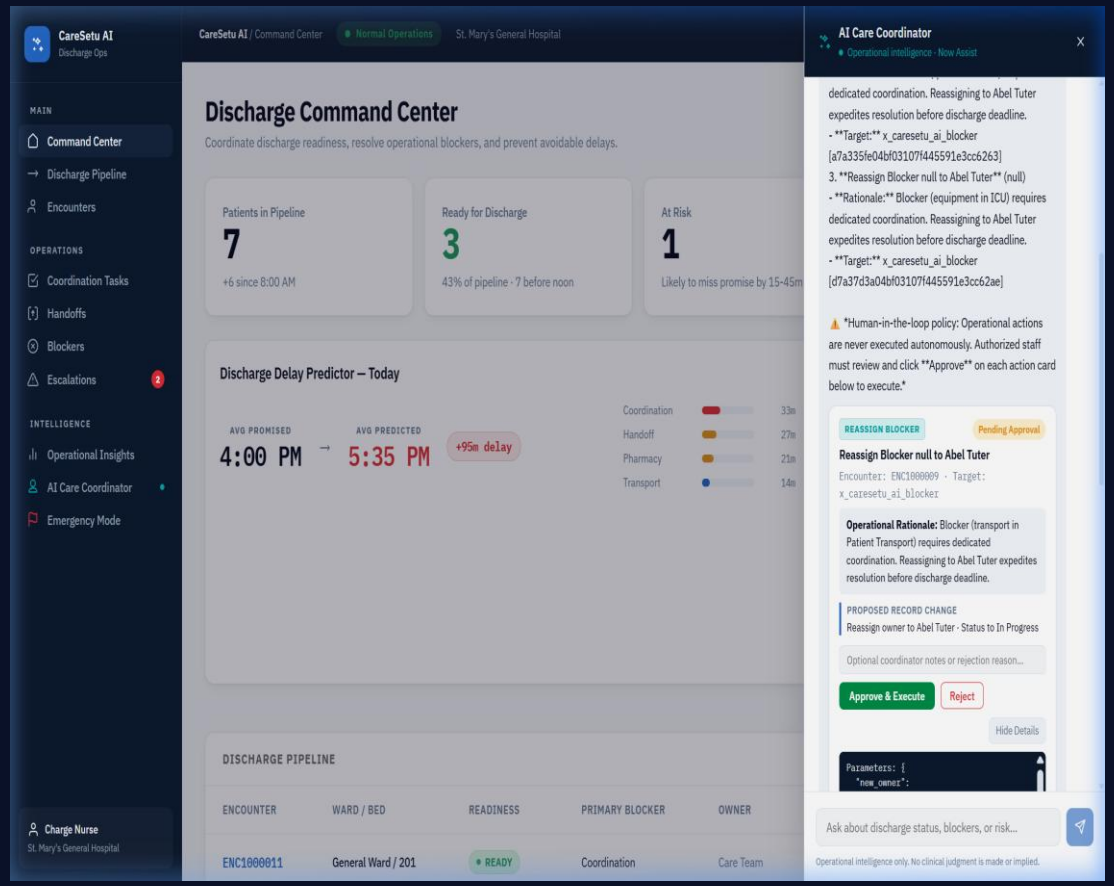
- Query: 'What is the current status of Bed 304?'
- Live Table Context: Aggregates records from encounter, blocker, tpa_case, and handoff tables
- Grounded Synthesis: Directly names Star Health TPA pre-auth as the primary bottleneck
- Root-Cause Attribution: Explains the exact +45m delay contribution
- Non-Clinical Boundary: Strictly addresses operational blockers; never offers medical advice
- Fluent Integration Ready: Now Assist Skill defined in code; in-scope deterministic engine guarantees zero downtime

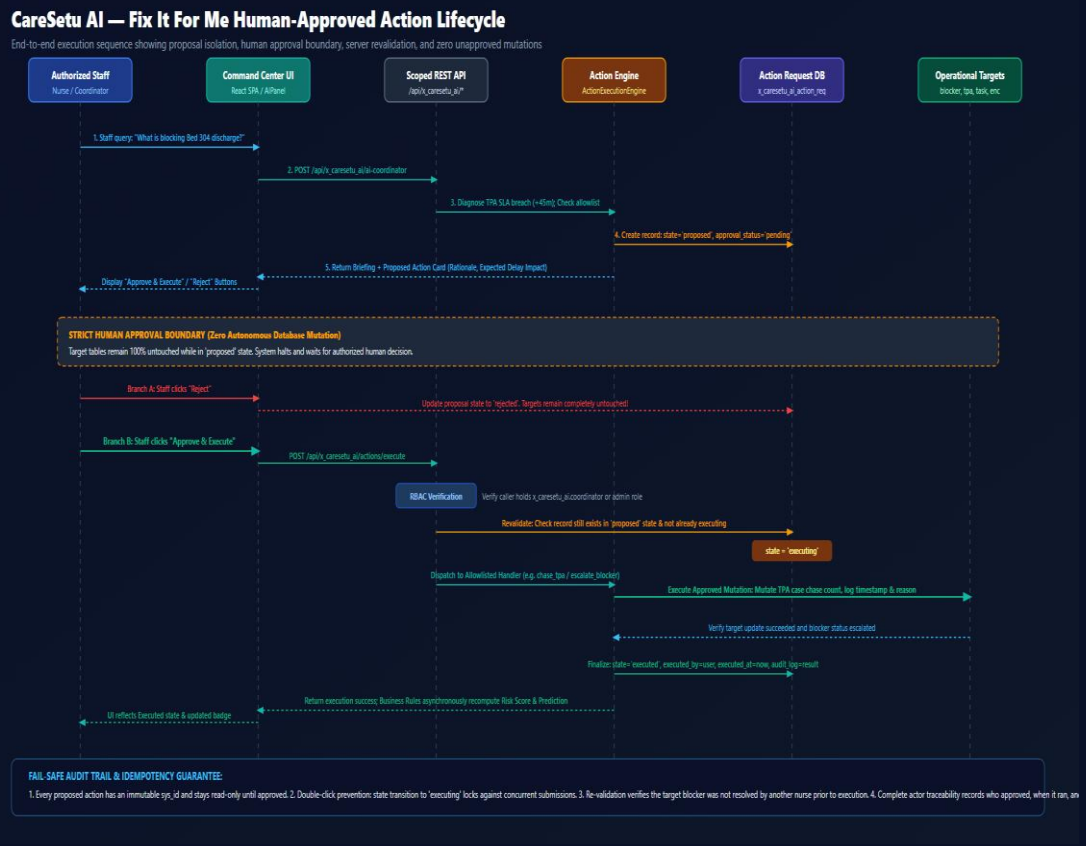
AI GOVERNANCE: The AI Care Coordinator never guesses or hallucinates: every sentence is verified against active ServiceNow records.

SAFE OPERATIONAL MUTATIONS

'Fix It For Me': Human-Approved Action Lifecycle

AI identifies bottlenecks and proposes allowlisted fixes. Authorized humans approve. Server validates and executes.





SAFETY PIPELINE: Proposal State Isolation -> Role Verification (coordinator/admin) -> Revalidation -> Allowlisted Execution -> Immutable Audit Trail

AUTOMATION & SYSTEM BOUNDARIES

Autonomous Schedulers & Honest Production Limits

Clear distinction between verified working engines and unconfigured external integrations.

Verified Implemented Engines

- TpaChaseEngine: Scheduled job runs every 15m; checks submission SLAs (15m/30m/45m intervals); logs chase attempts and updates follow-up counters.
- HandoffEngine: Manages shift transitions; monitors 30-minute ACK SLA; auto-generates reminders and escalates unacknowledged handoffs into blockers.
- OperationalRiskEngine: Async Business Rules recompute 0–100 risk score on any blocker, task, milestone, or handoff update.
- DischargePredictionEngine: Dynamically recalculates predicted discharge time and delay minutes using category deduplication.
- Escalation Synchronization: Automatically creates and deactivates x_caresetu_ai_escalation records at risk >= 60 threshold.

Honest Production Boundaries

- Authoritative Hospital EHR Telemetry: Not integrated; demo operates strictly on scoped synthetic records (ENC1000007).
- External Insurer Live Clearinghouse: External payer API not configured; TPA chaser executes internal operational tracking lifecycle.
- Emergency Operations Module: Surge view and rapid blocker triage are fully implemented in UI and data, but hospital-wide bed census telemetry is unconfigured.
- Now Assist Platform Activation: Fluent skill config is written; in-scope grounded aggregator is active as the reliable fallback.
- Strict Non-Clinical Scope: CareSetu never diagnoses illness, never recommends drug dosages, and never overrides physician authority.

TRANSPARENCY: Engineering integrity: We explicitly document working capabilities vs. unconfigured external hospital integrations.

IMPACT & VISION

Restoring Time to Patients and Beds

The right care, in the right bed, at the right hour.

Operational Impact

- Target 45–60 minute reduction in average discharge turnaround time.
- Unblocks ~1,000+ bed-hours per month in a 400-bed facility.
- Reduces hallway boarding and emergency department gridlock.
- Cuts nurse administrative chasing by an estimated 50%.

Enterprise Healthcare Guardrails

- Role-Based Access Control: admin, coordinator, and user segregation.
- Human approval boundary: zero autonomous mutation.
- Full idempotency & audit logging on every action execution.
- HIPAA & synthetic testing: zero real patient PHI.

The Road Ahead

- FHIR R4 & HL7 v2 connectors for enterprise EHR synchronization.
- External insurer portal webhooks for automated pre-auth confirmation.
- Multi-hospital network capacity balancing across regional health systems.
- Native Now Assist GenAI activation for ambient ward briefing.

THE MISSION: When hospitals run on connected operational workflows, patients like Nanaji go home the hour they are healed — not hours later.